Matplotlib & Seaborn: Detailed Summary

Detailed Summary: Matplotlib & Seaborn

1. Session Focus

- Concept of data visualization.
- Importance of Matplotlib for customizable plots.
- Integration with NumPy and Pandas.
- Importance of Seaborn for high-level, attractive statistical graphics.
- Common plot types: line, bar, scatter, pie, box, violin, heatmap, KDE.

2. Matplotlib Basics

- Installation: bash   pip install matplotlib   conda install matplotlib
- Import: python   import matplotlib.pyplot as plt
- Figure Components:
- Plotting area
- Axis labels
- Title
- Legend
- Ticks

Example: Simple Line Plot

```
python
x = [1,2,3,4,5]
y = [2,3,5,7,11]
plt.plot(x, y, label="Prime Numbers")
plt.title("Simple Line Plot")
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.legend()
plt.show()
```

3. Customization in Matplotlib

- Markers: symbols like o (circle), * (star), s (square).
- Colors: abbreviations ('r', 'g', 'b') or names ('red', 'blue') or hex codes ('#FF5733').
- Line Style: 'solid', 'dashed', 'dotted', 'dashdot'.
- Line Width: default = 1 pixel, can be increased.

Example: Weight vs Height

```
python
plt.plot(weight, height, color='green', marker='*', markersize=10,
         linestyle='--', linewidth=2, label='Avg Weight vs Height')
plt.title("Average Weight vs Height")
plt.xlabel("Weight (kg)")
plt.ylabel("Height (cm)")
plt.legend()
plt.grid()
```

plt.show()

4. Integration with NumPy & Pandas

- NumPy: Efficient array handling for plotting.   python   import numpy as np   x = np.array([0,1,2,3,4,5])   y = x**2   plt.plot(x,
- Pandas: Direct plotting from DataFrames.
- Example: Gender distribution pie chart from Customers.csv.
- Example: City-wise bar chart for top 10 cities.

5. Plot Types & Their Usage

Line Plot

- Purpose: Show trends or changes over time/continuous variables.
- Usage: Monthly spending patterns, growth of sales, stock prices.
- Example: python   plt.plot(months, sales, marker='o')

Bar Plot

- Purpose: Compare categories side by side.
- Usage: Top 10 cities by customers, average cost of products by company.
- Example: python   sns.barplot(x='day', y='tip', data=tips, estimator='mean')

Pie Chart

- Purpose: Show proportions of categories in a whole.
- Usage: Gender distribution of customers, market share of companies.
- Example: python   plt.pie(gender_counts, labels=gender_counts.index, autopct='%1.1f%%')

Scatter Plot

- Purpose: Show relationships between two continuous variables.
- Usage: Total bill vs tip, height vs weight.
- Example: python   sns.scatterplot(x='total_bill', y='tip', data=tips, hue='sex')

Box Plot

- Purpose: Show distribution, median, quartiles, and outliers.
- Usage: Distribution of paid amounts, tips by day.
- Example: python   sns.boxplot(x='day', y='tip', data=tips)

Violin Plot

- Purpose: Combines box plot + KDE (distribution + summary).
- Usage: Compare distributions of tips across days.
- Example: python   sns.violinplot(x='day', y='tip', data=tips)

Heatmap

- Purpose: Show correlations or intensity values in a matrix.
- Usage: Correlation between numeric variables (e.g., total_bill, tip, size).
- Example: python   sns.heatmap(df.corr(), annot=True, cmap='coolwarm')

KDE Plot (Kernel Density Estimate)

- Purpose: Show smooth distribution curve of a variable.
- Usage: Distribution of total bill by time of day, compare density of continuous variables.
- Example: python   sns.kdeplot(x='total_bill', data=tips, hue='time', fill=True)

Grouped Bar Plot

- Purpose: Compare multiple categories simultaneously.
- Usage: Average tip by day and time (Lunch vs Dinner).
- Example: python   sns.barplot(x='day', y='tip', hue='time', data=tips)

6. Seaborn Advantages

- Built-in themes and color palettes.
- Concise syntax for complex plots.
- Specialized functions for statistical relationships.
- Works seamlessly with Pandas DataFrames.
- Covers wide range: line plots, bar plots, scatter plots, violin plots, heatmaps, pairplots.

7. Key Takeaways

- Matplotlib: Flexible, customizable, integrates with NumPy/Pandas.
- Seaborn: High-level, aesthetically pleasing, simplifies complex plots.
- Use Matplotlib for fine control, Seaborn for quick, attractive statistical graphics.
- Together, they form the backbone of Python data visualization.

8. Why This Matters

- Visualization makes data accessible and understandable.
- Helps identify patterns, trends, and outliers.
- Essential for EDA (Exploratory Data Analysis), reporting, and machine learning workflows.
- Strong visualization skills = better communication of insights.